Question 1		b	c	d	e	f
Question 2		b	c	d	e	f
Question 3	a		c	d	e	f
Question 4	a	b		d	e	f
Question 5		b	c	d	e	f
Question 6		b	c	d	e	f
Question 7		b	c	d	e	f
Question 8	a		c	d	e	f
Question 9		b	c	d	e	f
Question 10	a	b	c		e	f
Question 11	a		c	d	e	f
Question 12	a	b	c		e	f
Question 13	a		c	d	e	f
Question 14	a	b	c		e	f
Question 15	a	b		d	e	f
Question 16	a	b	c		e	f
Question 17		b	c	d	e	f
Question 18		b	c	d	e	f
Question 19		b	c	d	e	f
Question 20	a		c	d	e	f
Question 21	a		c	d	e	f
Question 22	a	b		d	e	f
Question 23		b	c	d	e	f
Question 24	a	b		d	e	f
Question 25	a		c	d	e	f

Solution to problem 1.

$$\begin{cases} x = v\cos(\theta + \emptyset)t \\ y = v\sin(\theta + \emptyset)t - \frac{g}{2}t^2 \end{cases}$$

from the equation of x , $t = \frac{x}{v\cos(\theta+\emptyset)}$

Putting into the equation y, $y = v\sin(\theta + \emptyset)\frac{x}{v\cos(\theta+\emptyset)} - \frac{g}{2}\left(\frac{x}{v\cos(\theta+\emptyset)}\right)^2$

On the inclined plane, it needs to satisfy: $\tan\theta = \frac{y}{x} \Rightarrow y = x\tan\theta$

Substituting into the previous equation,

$$x\tan\theta = v\sin(\theta + \emptyset)\frac{x}{v\cos(\theta+\emptyset)} - \frac{gx^2}{2v^2\cos(\theta+\emptyset)^2}$$

$\Rightarrow$ *Rearranging,* $x = \frac{2}{g}v^2\cos(\theta + \emptyset)^2(\tan(\theta + \emptyset) - \tan\theta)$

$$= \frac{2}{g}v^2(\cos(\theta + \emptyset)\sin(\theta + \emptyset) - \tan\theta\cos(\theta + \emptyset)^2$$

$$= \frac{v^2}{g}(\sin(2(\theta + \emptyset)) - 2\tan\theta\cos(\theta + \emptyset)^2)$$

To maximize the range of the ball,

$$\frac{dx}{d\emptyset} = 0 \Rightarrow \frac{v^2}{g}2\big(\cos(2\theta + \emptyset) + 4\tan\theta\cos(\theta + \emptyset)\sin(\theta + \emptyset)\big) = 0$$

$$\Rightarrow \cos(2(\theta + \emptyset) + \tan\theta\sin(2(\theta + \emptyset)) = 0$$

$\Rightarrow \tan\theta = -\cot\big(2(\theta + \emptyset)\big) \Rightarrow \cot\left(\frac{\pi}{2} + \theta\right) = \cot\big(2(\theta + \emptyset)\big) \Rightarrow 2(\theta + \emptyset) = \frac{\pi}{2} + \theta$

$\therefore \emptyset = \frac{\pi}{4} - \frac{\theta}{2}$

Solution to problem 2.

f= 1/((1.33 -1)(1/0.0325+1/0.0325)) = 49 mm

The distance from the place where the bottle touches the wood to the line of burn L is given by

$L = SQR(f^2 - r^2) = 0.0375$

The angle of the sun above the horizontal is asin(r/f)

angle 41°

as the distance to the sun is much larger than distance to the image we have

$d_i = f$

$h_i/f = h/d$

$h_i = f\ h/d = 49.2/110 = 0.45$ *mm*

The energy illuminating the bottle will be condensed to the size of the image:

$1\ kW/m^2 * 32.5/0.45 = 72\ kW/m^2$

As the conduction is negligible all the energy hitting the area A of the wood wood has to be radiated back

$73 kW/m^2 A = A\ \sigma\ (T^4 - 300^4)$

Area cancels

$T^4 = 73\ 000/5.7\ 10^{-8} + 300^4$

$T = 1062\ K = 789°C$

So the wood will ignite.

One could also say that the power per unit area needed to get to 500°C is:

$P/A = \sigma\ (773^4 - 300^4) = 20\ kW/m^2$

Which is less than 73 kW/m^2 *so the wood will ignite.*

Solution to problem 3.

a)By energy conservation,

$E_p + E_\gamma = E_\Delta$

By momentum conservation we have

$Pp - P\gamma = P_\Delta$

$E_\Delta^2 - (P_\Delta c)^2 = (m_\Delta c^2)^2$

$(E_p+E_\gamma)^2 - c^2(P_p-P_\gamma)^2 = (m_\Delta c^2)^2$

$(E_p^2 - c_2 P_p^2) + (E_\gamma^2 - c^2 p_\gamma^2) + 2(EpE_\gamma+c^2P_pP_\gamma) = (m_\Delta c^2)^2$

$E_\gamma = P_\gamma c$

$(E_p^2 - c_2 P_p^2) + 2 (E_pE_\gamma + c^2 P_pP_\gamma) = (m_\Delta c^2)^2$

$E_p + c\ P_p = \frac{m_\Delta^2 c^4 - m_p^2 c^4}{2E_\gamma}$

$Ep^2 = (P_pc)^2 + (m_pc^2)^2$

$P_p >> m_pc^2$

$E_p = cP_p$

$E_p = \frac{m_\Delta^2 c^4 - m_p^2 c^4}{4E_\gamma}$

b)$E_\gamma = hc/\lambda$

$E_\gamma = 1.87 \times 10^{-22}\ J$

$E_p = 21.8\ J = 1.3 \times 10^{20}\ eV$

c)$E_p + E_\gamma = E_\Delta$

$P_p - P_\gamma \cos\theta = P_\Delta \cos\theta'$

$P_\gamma \sin\theta = P_\Delta \sin\theta'$

Put together and set c=1 in order to make the math easier.

$m_\Delta^2 = E_\Delta^2 - P_\Delta^2 = (E_p + E_\gamma)^2 - (P_p - P_\gamma \cos\theta)^2 - (P_\gamma \sin\theta)^2$

$E_p^2 - P_p^2 + E_\gamma^2 - P_p^2 + 2(E_p E_\gamma + P_p P_\gamma \cos\theta)$

$E_p + P_p \cos\theta = (m_\Delta^2 - m_p^2)/2E_\gamma$

$E_p = (m_\Delta^2 - m_p^2) / 2E_\gamma (1+\cos\theta) \quad > E_p$ *from part a*

So, if this photon were to collide with the proton at a different angle, energy of the photon would be larger!